

The Choice of Normalization Influences Shrinkage in Regularized Regression

Johan Larsson^{1,2,*} Jonas Wallin²

¹Department of Mathematical Sciences, University of Copenhagen, Denmark

²The Department of Statistics, Lund University, Sweden

*johan@jolars.co



The takeaway

Normalization is not neutral preprocessing. For **binary features**, the *class balance* (proportion of ones) leaks into your lasso, ridge, and elastic net coefficients—and *which* normalization you pick decides whether it does.

The problem

Regularized models are scale-sensitive, so we *normalize* (center and scale) features first. But there are many ways to do this, and for binary features there is no obvious choice. Despite this, the effect of that choice has gone unstudied.

Below: two binary features, both with true coefficient 1, differing only in class balance (0.5 vs. 0.9). The normalization type alone flips which feature looks more important.

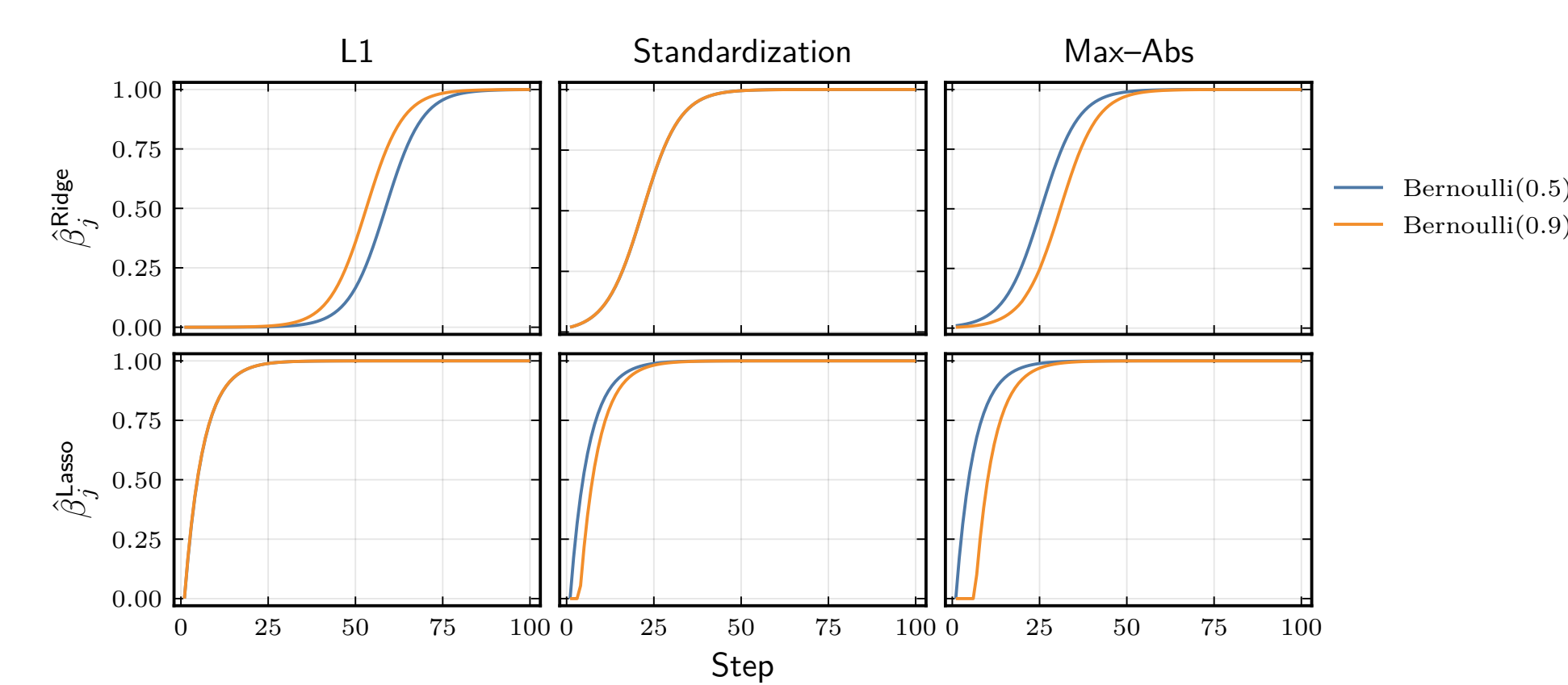


Figure 1: Noiseless lasso and ridge paths for two binary features. L_1 /standardization paths overlap; max-abs disagrees.

Setup

We study the elastic net (Zou & Hastie, 2005),

$$\frac{1}{2} \|\mathbf{y} - \beta_0 - \tilde{\mathbf{X}}\boldsymbol{\beta}\|_2^2 + \lambda_1 \|\boldsymbol{\beta}\|_1 + \frac{\lambda_2}{2} \|\boldsymbol{\beta}\|_2^2,$$

which gives the **lasso** ($\lambda_2 = 0$) and **ridge** ($\lambda_1 = 0$) as special cases. Each binary feature $\mathbf{x}_j$ is centered and scaled by $s_j = (q_j - q_j^2)^\delta$, $\delta \geq 0$, covering no scaling ($\delta = 0$), standard deviation ($\delta = \frac{1}{2}$), and variance ($\delta = 1$).

Class balance enters the estimate

With orthogonal features and no noise, the estimate on the original scale is

$$\hat{\beta}_j = \frac{\text{ST}_{\lambda_1} \left(\frac{\beta_j^* n (q_j - q_j^2)}{s_j} \right)}{s_j \left(\frac{n (q_j - q_j^2)}{s_j^2} + \lambda_2 \right)}.$$

The feature variance $q_j - q_j^2$ couples class balance to the estimate, mediated entirely by the scale s_j . The right δ makes it cancel:

Penalty	Cancelling scale δ
Lasso	variance 1
Ridge	standard dev. $\frac{1}{2}$
Elastic net	none exists —

For the elastic net, *no* s_j removes the bias—instead weight the *penalty* ($u_j = v_j = (q_j - q_j^2)^\omega$, $\omega = 1$).

Selection follows δ

Lower δ means imbalanced features are far less likely to be selected; as $q \rightarrow 1$, selection probability $\rightarrow 0$ for $\delta < \frac{1}{2}$ and $\rightarrow 1$ for $\delta > \frac{1}{2}$.

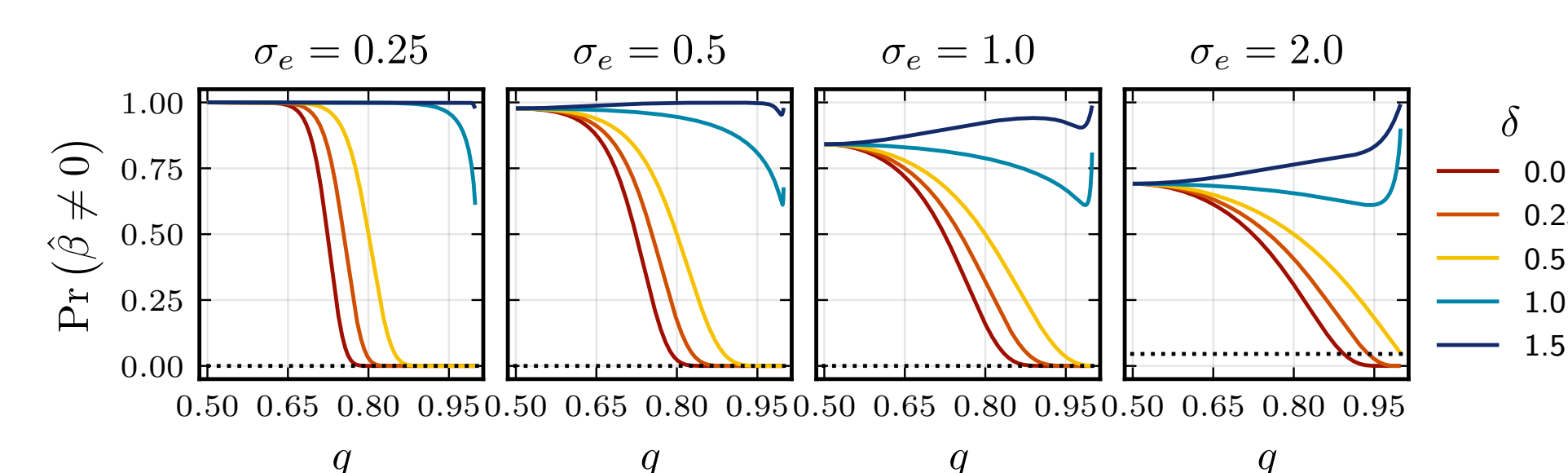


Figure 2: Lasso selection probability vs. class balance q for several δ . Dotted line: asymptotic limit at $\delta = \frac{1}{2}$.

A bias–variance trade-off

Larger δ removes class-balance bias but inflates variance, so the best δ depends on the noise level—which suggests tuning it.

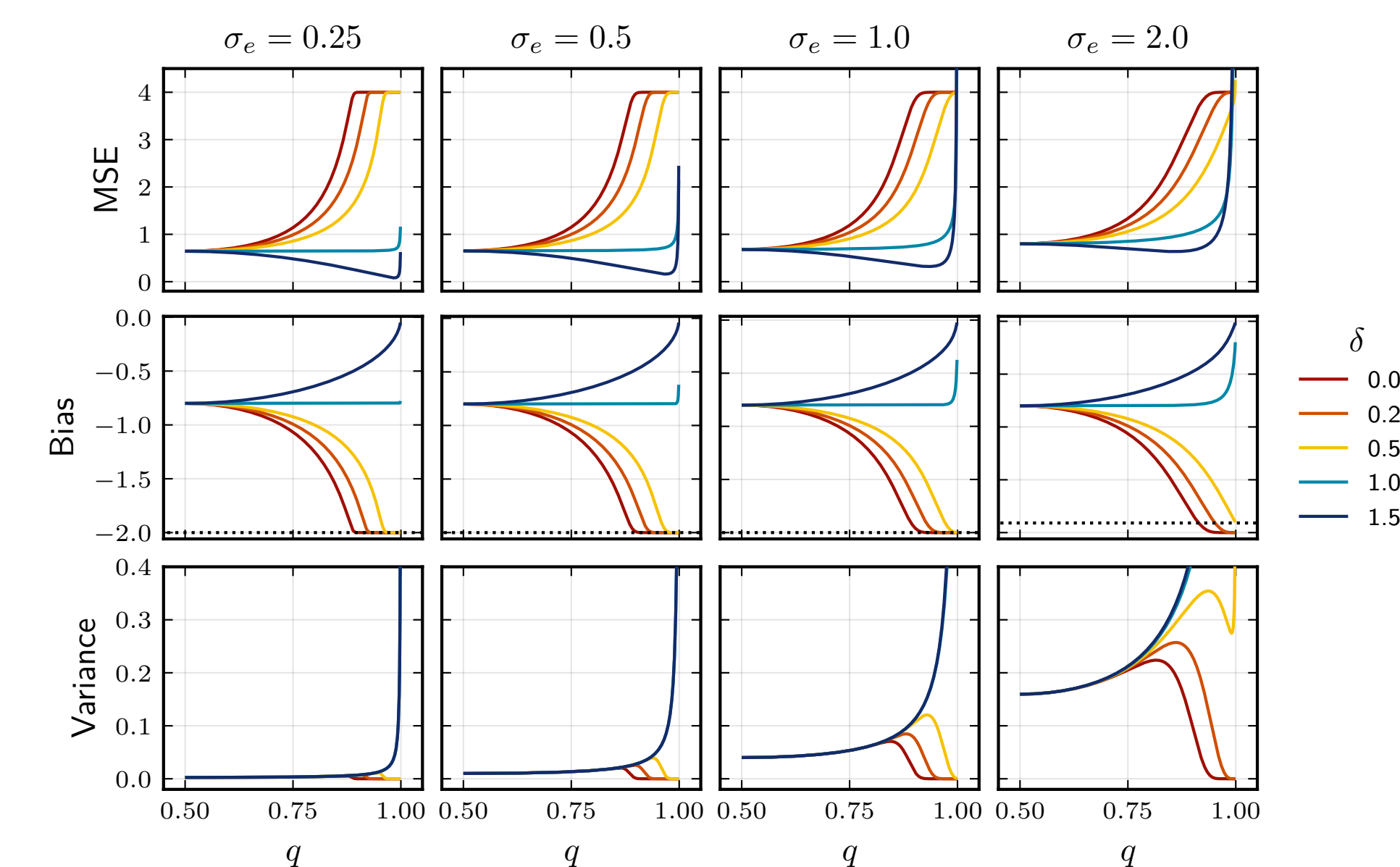
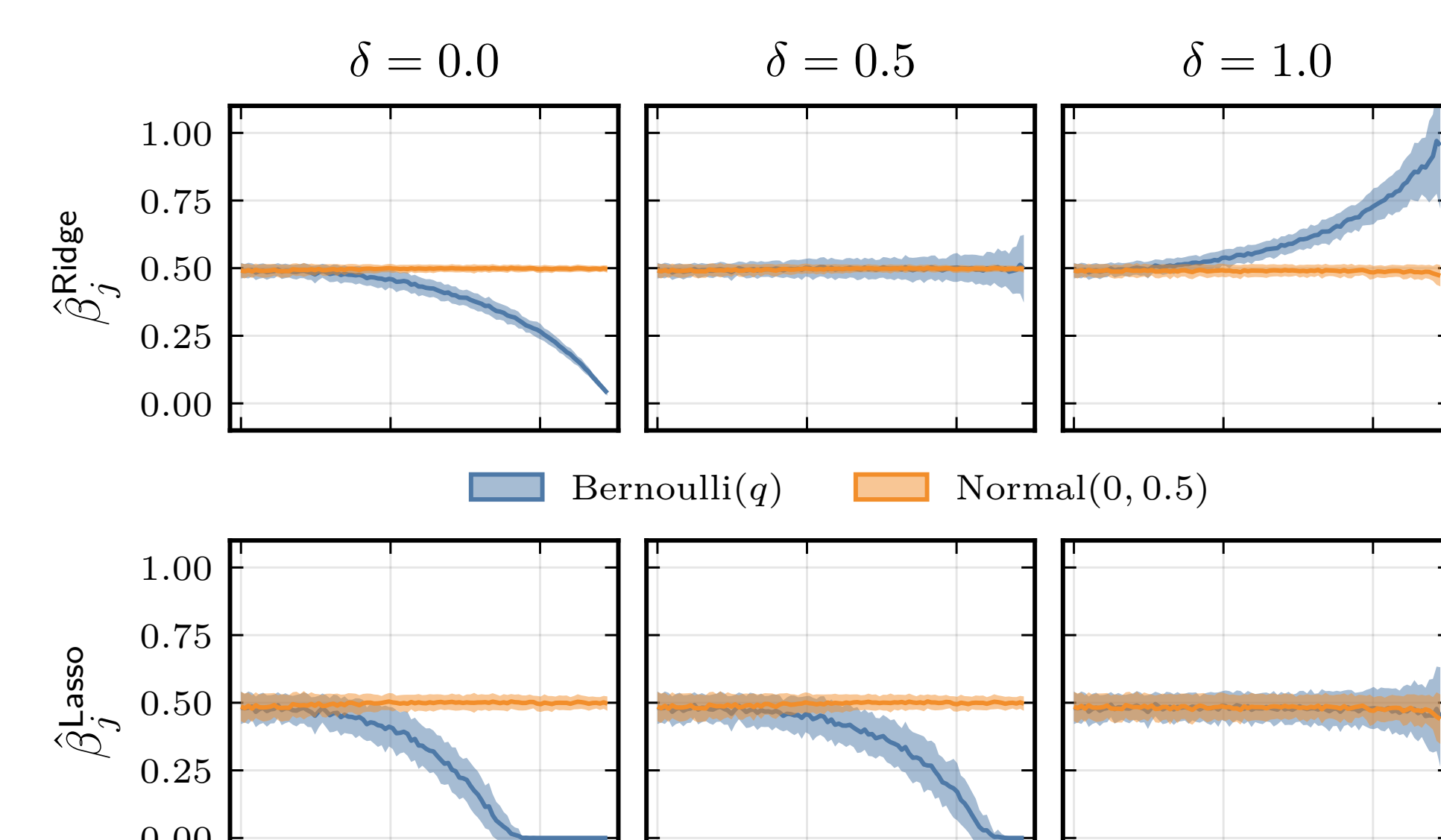


Figure 3: Bias, variance, and MSE for a one-dimensional lasso across class balance q , noise σ_ϵ , and scaling δ .

Mixed and interaction features

With both binary and continuous features, normalization implicitly sets their *relative* penalization (via a comparability constant κ (Gelman, 2008))—even at perfect balance. The lasso needs variance scaling, ridge needs standard-deviation scaling, to keep effects comparable.



Recommendations

Selection If missing a feature is costly, scale by *variance* ($\delta = 1$).

Prediction Standard deviation or no scaling; better, *tune* δ as a hyperparameter.

Elastic net Weight the *penalty* ($\omega = 1$), not the features.

Mixed/interactions Center before interacting; scale interactions by the product of parent scales.

Why it matters

The two most common defaults are *both* biased: standardization (the glmnet default; Friedman et al., 2010) and no scaling (recommended for sparse data in scikit-learn). The class-balance effect is therefore likely pervasive in published analyses. Our theory assumes orthogonality, but the experiments show the findings hold well beyond it.

References

- Zou & Hastie (2005). Regularization and variable selection via the elastic net. *JRSS-B*.
- Gelman (2008). Scaling regression inputs by dividing by two standard deviations. *Statistics in Medicine*.
- Friedman, Hastie & Tibshirani (2010). Regularization paths for generalized linear models via coordinate descent. *JSS*.
- Bien, Taylor & Tibshirani (2013). A lasso for hierarchical interactions. *Annals of Statistics*.
- Lim & Hastie (2015). Learning interactions via hierarchical group-lasso regularization. *JCGS*.

Contact

E-mail johan@jolars.co

Homepage <https://jolars.co>

Mastodon [@jolars@mastodon.social](https://jolars@mastodon.social)

BlueSky [@jolars.co](https://jolars.co)

